



EDUCACIÓN
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TECNOLÓGICO
NACIONAL DE MÉXICO



Instituto Tecnológico de San Juan del Río



TOPICOS DE CIBERSEGURIDAD

VALDES ARTEAGA LEONARDO

Hardening de Gnu-Linux_NARANJA

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PERIODO Enero-Junio

INTRODUCCIÓN

El hardening es el proceso de fortalecer la seguridad de un sistema operativo para reducir vulnerabilidades y protegerlo contra accesos no autorizados, ataques y robo de información.

En este manual se realizará el aseguramiento de un servidor GNU/Linux Ubuntu mediante:

- Protección SSH
- Configuración de firewall
- Administración segura de usuarios
- Protección de bases de datos
- Autenticación en dos pasos
- Respallos automáticos
- Prevención de ataques

OBJETIVO

Asegurar un servidor Ubuntu para minimizar riesgos de seguridad y proteger servicios críticos mediante configuraciones seguras.

1. ACTUALIZAR EL SISTEMA

¿Para qué sirve?

Actualizar el sistema corrige:

- Vulnerabilidades
- Errores de seguridad
- Fallos del sistema
- Bugs de paquetes

Mantener actualizado el servidor es la primera medida de seguridad.

Paso 1. Actualizar lista de repositorios



sudo apt update

```
eqnaranja:~#  
eqnaranja:~# sudo apt update  
Hit:1 http://mirrors.digitalocean.com/ubuntu noble InRelease  
Hit:2 https://repos-droplet.digitalocean.com/apt/droplet-agent main InRelease  
Hit:3 http://mirrors.digitalocean.com/ubuntu noble-updates InRelease  
Hit:4 http://mirrors.digitalocean.com/ubuntu noble-backports InRelease  
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1704 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1192 kB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [230 kB]  
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [9000 B]  
Fetched 3261 kB in 11s (288 kB/s)  
Reading package lists... 27%
```

Explicación del comando

Parte Función

sudo Ejecuta el comando como administrador

apt Administrador de paquetes de Ubuntu

update Descarga lista de paquetes disponibles

¿Qué hace exactamente?

Ubuntu consulta sus servidores oficiales y obtiene información de:

- nuevas versiones
- actualizaciones
- parches de seguridad

Paso 2. Actualizar paquetes instalados

sudo apt upgrade -y

```
eqnaranja:~#  
eqnaranja:~# sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages were automatically installed and are no longer required:  
  libcgi-fast-perl libcgi-pm-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl  
  libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl  
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic  
  mecab-ipadic-utf8 mecab-utils  
Use 'sudo apt autoremove' to remove them.  
Get another security update through Ubuntu Pro with 'esm-apps' enabled:  
  python3-wheel  
Learn more about Ubuntu Pro at https://ubuntu.com/pro  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
eqnaranja:~# |
```

Explicación

Parte Función

upgrade Instala actualizaciones disponibles

-y Acepta automáticamente

¿Qué hace?

Actualiza programas ya instalados:

- OpenSSH
- Kernel
- Librerías
- Servicios

Paso 3. Eliminar paquetes innecesarios

sudo apt autoremove -y

```
eqnaranja:~#  
eqnaranja:~# sudo apt autoremove -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages will be REMOVED:  
  libcgi-fast-perl libcgi-pm-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl  
  libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl  
  libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl liburi-perl mecab-ipadic  
  mecab-ipadic-utf8 mecab-utils  
0 upgraded, 0 newly installed, 21 to remove and 0 not upgraded.  
After this operation, 58.1 MB disk space will be freed.  
(Reading database ... 110926 files and directories currently installed.)  
Removing libcgi-fast-perl (1:2.17-1) ...  
Removing libhtml-template-perl (2.97-2) ...  
Removing libcgi-pm-perl (4.63-1) ...  
Removing libhttp-message-perl (6.45-1ubuntu1) ...
```

¿Para qué sirve?

Elimina paquetes viejos que ya no se usan y que pueden representar riesgos.

2. CREAR USUARIO ADMINISTRADOR

¿Por qué NO usar root?

El usuario root tiene control total del sistema.

Si alguien roba esa cuenta:

- controla todo el servidor
- puede borrar información
- instalar malware

- robar datos

Por eso se crea un usuario administrador secundario.

Paso 1. Crear usuario nuevo

sudo adduser seguridad

```
eqnaranja:~# sudo adduser seguridad
info: Adding user `seguridad' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `seguridad' (1000) ...
info: Adding new user `seguridad' (1000) with group `seguridad (1000)' ...
info: Creating home directory `/home/seguridad' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for seguridad
Enter the new value, or press ENTER for the default
  Full Name []: naranja
  Room Number []: eqnaranja
  Work Phone []: 554944065
  Home Phone []: 4272247835
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `seguridad' to supplemental / extra groups `users' ...
info: Adding user `seguridad' to group `users' ...
eqnaranja:~# |
```

Explicación

Parte Función

adduser Crea usuarios

seguridad Nombre del usuario

¿Qué pedirá el sistema?

- Contraseña
- Nombre completo
- Información opcional

Paso 2. Dar permisos administrativos

sudo usermod -aG sudo seguridad

```
eqnaranja:~#  
eqnaranja:~# sudo usermod -aG sudo seguridad  
eqnaranja:~# |
```

Explicación COMPLETA

Parte	Función
usermod	Modifica usuarios
-a	Agrega
-G	Agrega a grupo
sudo	Grupo administrador
seguridad	Usuario

¿Qué hace?

Permite que el usuario ejecute comandos administrativos usando sudo.

Paso 3. Verificar grupos

groups seguridad

```
eqnaranja:~#  
eqnaranja:~# groups seguridad  
seguridad : seguridad sudo users  
eqnaranja:~#  
eqnaranja:~# |
```

¿Qué hace?

Muestra grupos a los que pertenece el usuario.

Debe aparecer:

Sudo

3. CAMBIAR EL PUERTO SSH

¿Qué es SSH?

SSH permite administrar el servidor remotamente.

Por defecto usa:

Puerto 22

Ese puerto es atacado constantemente por bots.

Objetivo

Cambiar el puerto reduce:

- ataques automáticos
- escaneos
- intentos de fuerza bruta

Paso 1. Abrir configuración SSH

```
sudo nano /etc/ssh/sshd_config
```

```
eqnaranja:~#  
eqnaranja:~# sudo nano /etc/ssh/sshd_config  
eqnaranja:~# |
```

Explicación

Parte	Función
nano	Editor de texto
/etc/ssh/sshd_config	Archivo de configuración SSH

Paso 2. Buscar esta línea

```
#Port 22
```

```
#  
#Port 22  
#AddressFamily any
```

Paso 3. Cambiar puerto

Ejemplo:

Port 1222

```
Port 1222|  
#AddressFamily any  
#ListenAddress 0.0.0.0  
#ListenAddress ::
```

¿Qué hace?

SSH dejará de escuchar en el puerto 22 y usará el 1222.

Paso 4. Guardar archivo

Teclas:

- CTRL + O
- ENTER
- CTRL + X

Paso 5. Reiniciar SSH

sudo systemctl restart ssh

```
eqnaranja:~#  
eqnaranja:~# sudo systemctl restart ssh  
eqnaranja:~# |
```

Explicación

Parte	Función
systemctl	Controla servicios
restart	Reinicia
ssh	Servicio SSH

¿Por qué es necesario?

Para aplicar cambios.

Paso 6. Verificar estado

sudo systemctl status ssh

```
eqnaranja:~# sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-05-28 17:01:56 UTC; 16s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 27858 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 27859 (sshd)
    Tasks: 1 (limit: 1110)
   Memory: 1.2M (peak: 1.4M)
      CPU: 28ms
   CGroup: /system.slice/ssh.service
           └─27859 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 28 17:01:55 eqnaranja systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 28 17:01:56 eqnaranja sshd[27859]: Server listening on 0.0.0.0 port 1222.
May 28 17:01:56 eqnaranja sshd[27859]: Server listening on :: port 1222.
May 28 17:01:56 eqnaranja systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
eqnaranja:~#
```

¿Qué muestra?

- si SSH funciona
- errores
- puerto activo

4. CONFIGURAR FIREWALL (UFW)

¿Qué es un firewall?

Controla:

- qué conexiones entran
- qué conexiones salen

Funciona como una puerta de seguridad.

Objetivo

Permitir únicamente tráfico autorizado.

Paso 1. Instalar UFW

```
sudo apt install ufw -y
```

```
eqnaranja:~# sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
eqnaranja:~# |
```

¿Qué es UFW?

Uncomplicated Firewall.

Firewall oficial de Ubuntu.

Paso 2. Permitir puerto SSH nuevo

```
sudo ufw allow 1222/tcp
```

```
eqnaranja:~# sudo ufw allow 1222/tcp  
Rule added  
Rule added (v6)  
eqnaranja:~# |
```

Explicación

Parte Función

allow Permite tráfico

2222 Puerto

tcp Protocolo

Paso 3. Permitir HTTP

```
sudo ufw allow 80/tcp
```

```
eqnaranja:~# sudo ufw allow 80/tcp  
Rules updated  
Rules updated (v6)  
eqnaranja:~# |
```

¿Para qué sirve?

Permite páginas web HTTP.

Paso 4. Permitir HTTPS

```
sudo ufw allow 443/tcp
```

```
eqnaranja:~#  
eqnaranja:~# sudo ufw allow 443/tcp  
Rules updated  
Rules updated (v6)  
eqnaranja:~#  
eqnaranja:~#
```

¿Para qué sirve?

Permite tráfico web seguro.

Paso 5. Activar firewall

```
sudo ufw enable  
eqnaranja:~#  
eqnaranja:~# sudo ufw enable  
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y  
Firewall is active and enabled on system startup  
eqnaranja:~#  
eqnaranja:~#
```

¿Qué hace?

Activa reglas del firewall.

Paso 6. Ver reglas

```
sudo ufw status numbered
```

```
eqnaranja:~# sudo ufw status numbered
Status: active

      To Action From
      --
[ 1] 2222/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 443/tcp ALLOW IN Anywhere
[ 4] 22/tcp DENY IN Anywhere
[ 5] 1222/tcp ALLOW IN Anywhere
[ 6] 2222/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 8] 443/tcp (v6) ALLOW IN Anywhere (v6)
[ 9] 22/tcp (v6) DENY IN Anywhere (v6)
[10] 1222/tcp (v6) ALLOW IN Anywhere (v6)

eqnaranja:~# |
```

Parte Función

`sudo` Ejecuta el comando como administrador

`ufw` Uncomplicated Firewall (firewall de Ubuntu)

`status` Muestra el estado y reglas

`numbered` Enumera las reglas con números

¿Qué muestra?

- reglas activas
- puertos permitidos
- puertos bloqueados

Paso 7. Bloquear puerto 22

`sudo ufw deny 22/tcp`

```
eqnaranja:~# sudo ufw deny 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
eqnaranja:~# |
```

¿Por qué?

Para impedir accesos por el puerto SSH original.

5. DESHABILITAR ROOT

¿Por qué?

Root es el objetivo principal de hackers.

Paso 1. Editar configuración SSH

```
sudo nano /etc/ssh/sshd_config
```

```
GNU nano 7.2 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
```

Paso 2. Buscar

```
PermitRootLogin yes
```

```
#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

Paso 3. Cambiar por

PermitRootLogin no

```
#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

¿Qué hace?

Bloquea acceso remoto SSH del usuario root.

Paso 4. Reiniciar SSH

sudo systemctl restart ssh

```
eqnaranja:~#
eqnaranja:~# sudo systemctl restart ssh
eqnaranja:~# |
```

Paso 5. Bloquear contraseña root

sudo passwd -l root

```
eqnaranja:~# sudo passwd -l root
passwd: password changed.
eqnaranja:~# |
```

Explicación

Parte Función

passwd Maneja contraseñas

-l Lock (bloquea)

root Usuario root

¿Qué hace?

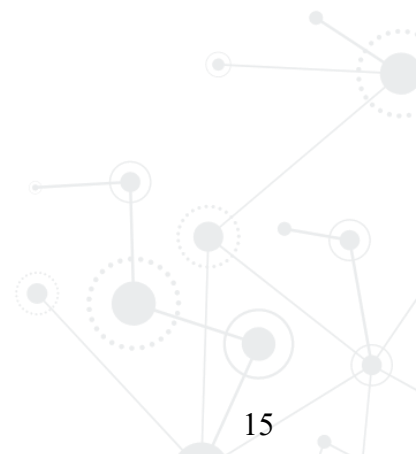
Deshabilita inicio de sesión del root.

Paso 6. Probar conexión nueva

Desde otra terminal:

```
ssh usuario@IP -p 1222
```

Si entra bien, ya quedó.



```
Windows PowerShell
seguridad@eqnaranja: ~
seguridad@eqnaranja: ~

PS C:\Users\maryp> ssh seguridad@159.203.33.12 -p 1222
seguridad@159.203.33.12's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu May 28 17:10:00 UTC 2026

System load:          0.0
Usage of /:           11.8% of 23.17GB
Memory usage:        23%
Swap usage:          0%
Processes:           109
Users logged in:      1
IPv4 address for eth0: 159.203.33.12
IPv4 address for eth0: 10.20.0.5
IPv6 address for eth0: 2604:a880:cad:d0:0:1:83f5:1001

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

seguridad@eqnaranja:~$ |
```

6. CLAVES SSH (AUTENTICACIÓN SEGURA)

¿Qué son?

SSH usa:

- clave pública
- clave privada

En vez de contraseña.

¿Por qué es más seguro?

Porque:

- no se puede adivinar fácilmente
- protege contra fuerza bruta
- usa criptografía

Paso 1. Generar claves

EN TU COMPUTADORA LOCAL:

```
ssh-keygen -t rsa -b 4096
```

```
PS C:\Users\maryp> ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (C:\Users\maryp\.ssh\id_rsa): y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in y
Your public key has been saved in y.pub
The key fingerprint is:
SHA256:YwzLw0v7UUFK6e0p6ZsYKp9EHRN3huPphXVAoSkJ/kg maryp@Mary-Perez
The key's randomart image is:
+---[RSA 4096]---+
|  . . o+Bo      |
|  . o =+B. .    |
|  E Bo=*..      |
|  . + B=+o.     |
|  o +.o+S.      |
|  . . +oo.      |
|  . o...        |
|  .. o +.o      |
|  o+ o.+        |
+---[SHA256]-----+
PS C:\Users\maryp> |
```

Explicación

Parte	Función
ssh-keygen	Genera claves SSH
-t rsa	Tipo RSA
-b 4096	Tamaño de clave

Paso 2. Copiar llave al servidor

```
type $env:USERPROFILE\.ssh\id_rsa.pub | ssh seguridad@159.203.33.12 -p 1222 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
```

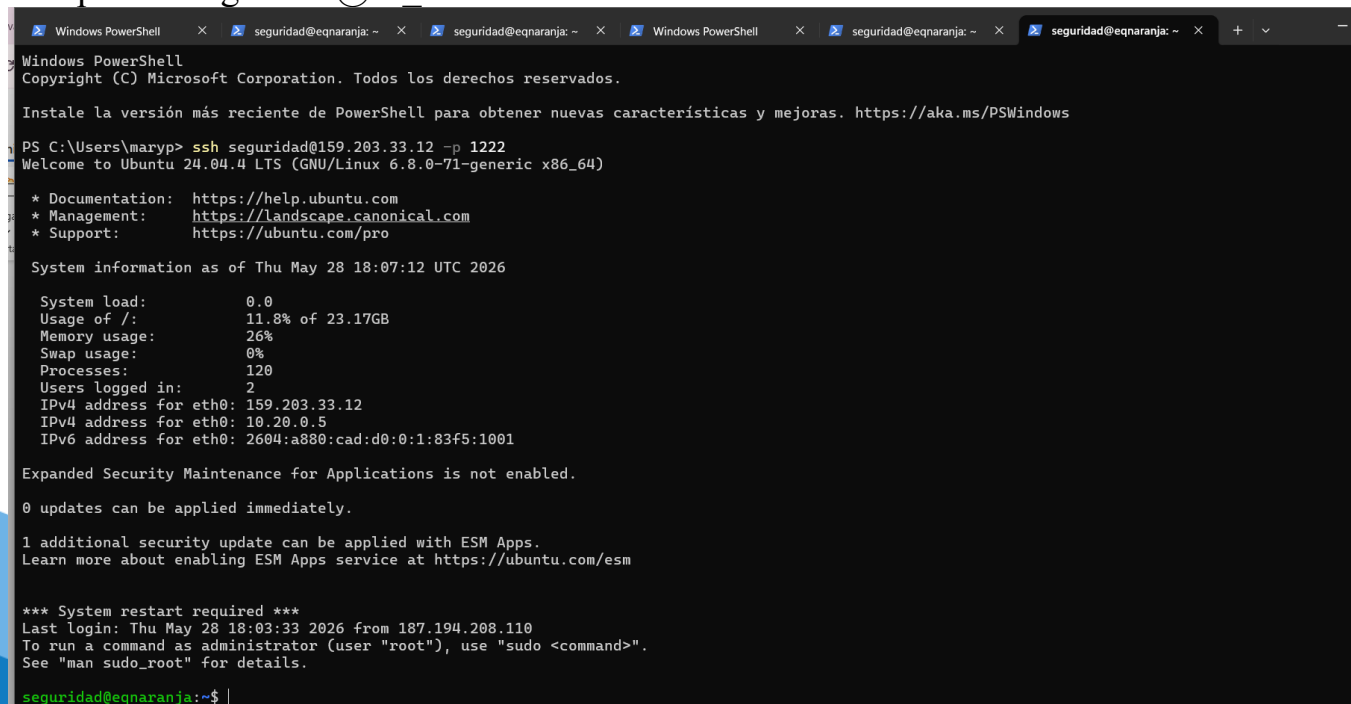
¿Qué hace?

Copia automáticamente la llave pública.

```
PS C:\Users\maryp> type $env:USERPROFILE\.ssh\id_rsa.pub | ssh seguridad@159.203.33.12 -p 1222 "mkdir -p
h && cat >> ~/.ssh/authorized_keys"
seguridad@159.203.33.12's password:
PS C:\Users\maryp> |
```

Paso 3. Entrar al servidor

```
ssh -p 1222 seguridad@IP_SERVIDOR
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. https://aka.ms/PSWindows

PS C:\Users\maryp> ssh seguridad@159.203.33.12 -p 1222
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu May 28 18:07:12 UTC 2026

System load:          0.0
Usage of /:            11.8% of 23.17GB
Memory usage:         26%
Swap usage:           0%
Processes:            120
Users logged in:       2
IPv4 address for eth0: 159.203.33.12
IPv4 address for eth0: 10.20.0.5
IPv6 address for eth0: 2604:a880:cad:d0:0:1:83f5:1001

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Thu May 28 18:03:33 2026 from 187.194.208.110
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

seguridad@eqnaranja:~$ |
```

7. DESHABILITAR CONTRASEÑAS SSH

Objetivo

Obligar uso de claves SSH.

Paso 1. Abrir configuración

`sudo nano /etc/ssh/sshd_config`

```
GNU nano 7.2 /etc/ssh/sshd_config
# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
```

Paso 2. Buscar

PasswordAuthentication yes

```
# Accepting forwarding no
# PermitTTY no
# ForceCommand cvs server

PasswordAuthentication yes
```

Paso 3. Cambiar por

PasswordAuthentication no

```
# PermitTTY no
# ForceCommand cvs server

PasswordAuthentication no|
```

¿Qué hace?

Impide login por contraseña.

Solo permitirá:

- llaves SSH

Paso 4. Reiniciar SSH

```
sudo systemctl restart ssh
```

```
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo systemctl restart ssh
seguridad@eqnaranja:~$ |
```

8. GOOGLE AUTHENTICATOR (2FA)

¿Qué es?

Autenticación en dos pasos.

El usuario necesita:

1. Llave SSH
2. Código temporal del celular

Beneficios

Aunque roben:

- contraseña

- llave SSH

NO podrán entrar sin el código temporal.

Paso 1. Instalar módulo

`sudo apt install libpam-google-authenticator -y`

```
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$ sudo apt install libpam-google-authenticator -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libqrencode4  
The following NEW packages will be installed:  
  libpam-google-authenticator libqrencode4  
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.  
Need to get 71.8 kB of archives.  
After this operation, 212 kB of additional disk space will be used.  
Get:1 http://mirrors.digitalocean.com/ubuntu noble/universe amd64 libqrencode4 amd64 4.1.1-1build2 [25.0 kB]  
Get:2 http://mirrors.digitalocean.com/ubuntu noble/universe amd64 libpam-google-authenticator amd64 20191231-2build1 [46.8 kB]  
Fetched 71.8 kB in 10s (7068 B/s)  
Selecting previously unselected package libqrencode4:amd64.  
(Reading database ... 110558 files and directories currently installed.)  
Preparing to unpack .../libqrencode4_4.1.1-1build2_amd64.deb ...  
Unpacking libqrencode4:amd64 (4.1.1-1build2) ...  
Selecting previously unselected package libpam-google-authenticator.  
Preparing to unpack .../libpam-google-authenticator_20191231-2build1_amd64.deb ...  
Unpacking libpam-google-authenticator (20191231-2build1) ...  
Setting up libqrencode4:amd64 (4.1.1-1build2) ...  
Setting up libpam-google-authenticator (20191231-2build1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Pending kernel upgrade!  
Running kernel version:  
  6.8.0-71-generic  
Diagnostics:  
  The currently running kernel version is not the expected kernel version 6.8.0-117-generic.  
  
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.  
  
Restarting services...
```

Paso 2. Ejecutar asistente

`google-authenticator`



Paso 3. Responder preguntas

Pregunta	Respuesta
----------	-----------

time-based tokens	y
-------------------	---

update file	y
-------------	---

disallow reuse	y
----------------	---

increase window	y
-----------------	---

rate limiting	y
---------------	---

Eso es la configuración de Google Authenticator para activar autenticación en dos pasos (2FA) en tu servidor SSH.

Cada parte sirve para esto:

- **Your new secret key**

Your new secret key is: ...

Es la clave secreta que se comparte entre:

- tu servidor Ubuntu
- la app Google Authenticator de tu celular

Gracias a esa clave ambos generan los mismos códigos temporales cada 30 segundos.

- **Enter code from app**

Enter code from app

Aquí escribiste el código de 6 dígitos que aparecía en la app del celular para comprobar que quedó vinculada correctamente.

- **Emergency scratch codes**

Your emergency scratch codes are:

Son códigos de emergencia de un solo uso.

Sirven si:

- pierdes el celular
- borras Google Authenticator
- cambias de teléfono

Con uno de esos códigos puedes entrar al servidor aunque no tengas la app.

Guárdalos en un lugar seguro.

- **Update .google_authenticator file**

Do you want me to update your `"/home/seguridad/.google_authenticator"` file?

Crea el archivo donde se guarda la configuración del 2FA para el usuario seguridad.

- **Disallow multiple uses**

Do you want to disallow multiple uses of the same authentication token?

Evita reutilizar el mismo código temporal más de una vez.

Beneficio:

- protege contra ataques “man in the middle”
- evita reutilización de tokens robados

- **Increase window**

Do you want to do so?

Esto amplía el tiempo permitido para aceptar códigos.

Tú respondiste y, entonces:

- el servidor acepta más códigos cercanos en tiempo
- ayuda si hay desfase de reloj entre servidor y celular

```
Your new secret key is: 7UIJCE5QB5WWG2732TQYD2SEI4
Enter code from app (-1 to skip): 843468
Code confirmed
Your emergency scratch codes are:
12179381
79700657
99858574
81559799
76201348

Do you want me to update your "/home/seguridad/.google_authenticator" file? (y/n) y

Do you want to disallow multiple uses of the same authentication
token? This restricts you to one login about every 30s, but it increases
your chances to notice or even prevent man-in-the-middle attacks (y/n) y

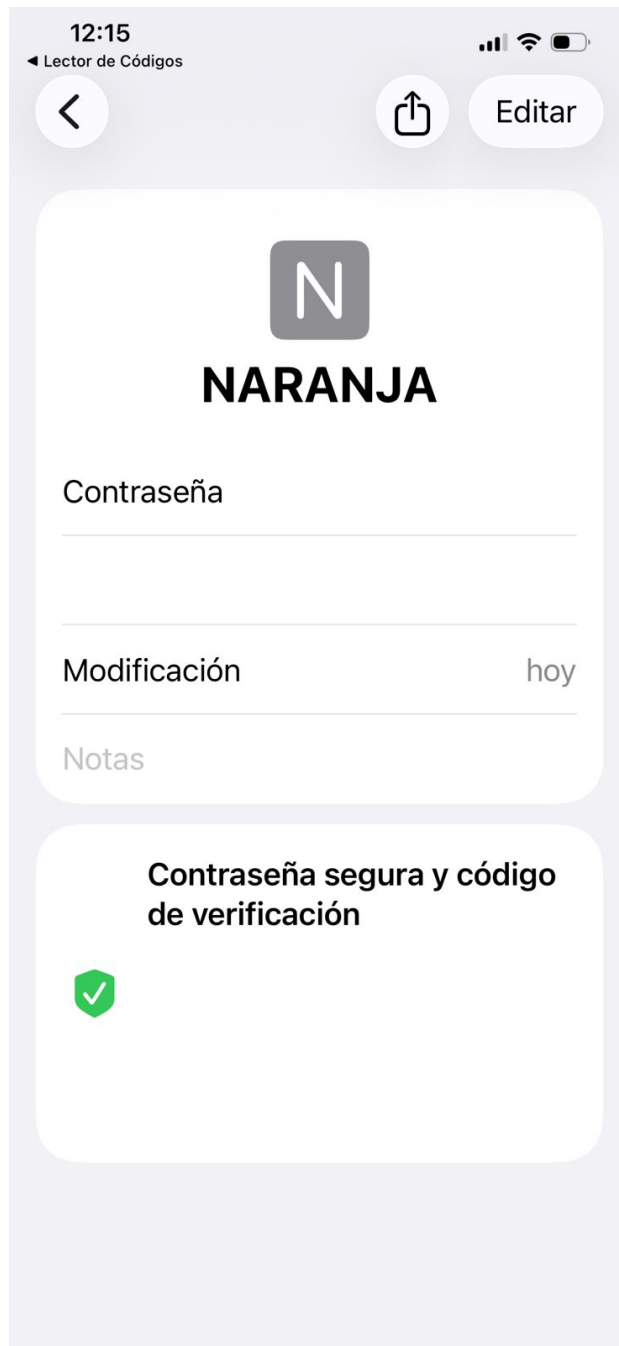
By default, a new token is generated every 30 seconds by the mobile app.
In order to compensate for possible time-skew between the client and the server,
we allow an extra token before and after the current time. This allows for a
time skew of up to 30 seconds between authentication server and client. If you
experience problems with poor time synchronization, you can increase the window
from its default size of 3 permitted codes (one previous code, the current
code, the next code) to 17 permitted codes (the 8 previous codes, the current
code, and the 8 next codes). This will permit for a time skew of up to 4 minutes
between client and server.
Do you want to do so? (y/n) y

If the computer that you are logging into isn't hardened against brute-force
login attempts, you can enable rate-limiting for the authentication module.
By default, this limits attackers to no more than 3 login attempts every 30s.
Do you want to enable rate-limiting? (y/n) y
seguridad@eqnaranja:~$ |
```

Paso 4. Escanear QR

Aplicación:

- Google Authenticator



Paso 5. Configurar PAM

```
sudo nano /etc/pam.d/sshd
```

```

GNU nano 7.2
# PAM configuration for the Secure Shell service

# Standard Un*x authentication.
@include common-auth

# Disallow non-root logins when /etc/nologin exists.
account    required    pam_nologin.so

# Uncomment and edit /etc/security/access.conf if you need to set complex
# access limits that are hard to express in sshd_config.
# account  required    pam_access.so

# Standard Un*x authorization.
@include common-account

# SELinux needs to be the first session rule. This ensures that any
# lingering context has been cleared. Without this it is possible that a
# module could execute code in the wrong domain.
session [success=ok ignore=ignore module_unknown=ignore default=bad]    pam_selinux.so close

# Set the loginuid process attribute.
session    required    pam_loginuid.so

# Create a new session keyring.
session    optional    pam_keyinit.so force revoke

# Standard Un*x session setup and teardown.
@include common-session

# Print the message of the day upon successful login.
# This includes a dynamically generated part from /run/motd.dynamic
# and a static (admin-editable) part from /etc/motd.
session    optional    pam_motd.so    motd=/run/motd.dynamic
session    optional    pam_motd.so    nouupdate

# Print the status of the user's mailbox upon successful login.
session    optional    pam_mail.so    standard noenv # [1]

# Set up user limits from /etc/security/limits.conf.
session    required    pam_limits.so

# Read environment variables from /etc/environment and
# /etc/security/pam_env.conf.
session    required    pam_env.so # [1]
# In Debian 4.0 (etch), locale-related environment variables were moved to
# /etc/default/locale, so read that as well.
session    required    pam_env.so    user_readenv=1 envfile=/etc/default/locale

# SELinux needs to intervene at login time to ensure that the process starts
# in the proper default security context. Only sessions which are intended
# to run in the user's context should be run after this.
session [success=ok ignore=ignore module_unknown=ignore default=bad]    pam_selinux.so open

# Standard Un*x password updating.
@include common-password

```

Agregar:

auth required pam_google_authenticator.so

```
# Configurar PAM
auth required pam_google_authenticator.so
```

Sirve para activar la autenticación en dos pasos (2FA) usando Google Authenticator en el login del sistema.

Paso 6. Configurar SSH

`sudo nano /etc/ssh/sshd_config`

```
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo nano /etc/ssh/sshd_config

GNU nano 7.2 /etc/ssh/sshd_config *
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
```

Agregar:

`KbdInteractiveAuthentication yes`

```
# Change to yes to enable challenge-response passwords (beware of
# some PAM modules and threads)
KbdInteractiveAuthentication yes
# Use the following options
```

`AuthenticationMethods publickey,keyboard-interactive`

```
AuthenticationMethods publickey,keyboard-interactive |
```

Esas líneas en `sshd_config` sirven para configurar cómo se autentican los usuarios en SSH y habilitar el segundo factor de autenticación.

La primer línea:

`KbdInteractiveAuthentication yes`

Sirve para permitir autenticación interactiva en SSH.

¿Qué significa “interactiva”?

Que el servidor puede hacer preguntas adicionales al usuario durante el inicio de sesión.

Por ejemplo:

Verification code:

Eso es exactamente lo que necesita Google Authenticator para pedir el código temporal del celular.

Si esa línea está en:

no

o deshabilitada:

- SSH NO podrá pedir códigos 2FA
- keyboard-interactive no funcionará
- aparecerán errores como el que te salió

¿Por qué es importante?

Porque esta línea **habilita** el segundo factor de autenticación.

Entonces el acceso SSH queda así:

1. Verifica tu llave SSH
2. SSH hace una pregunta interactiva:

Verification code:

3. Escribes el código del celular
4. Si es correcto → acceso permitido

Relación con esta línea:

AuthenticationMethods publickey,keyboard-interactive

Ahí le dices a SSH:

- primero exige llave SSH (publickey)
- después exige autenticación interactiva (keyboard-interactive)

Y KbdInteractiveAuthentication yes es lo que permite que esa segunda parte funcione.

La segunda línea:

AuthenticationMethods publickey,keyboard-interactive

Define QUÉ métodos de autenticación son obligatorios.

Se interpreta así:

- publickey

Primero exige:

- llave SSH privada
- keyboard-interactive

Después exige:

- código de Google Authenticator

Entonces el login queda:

1. Llave SSH correcta
2. Código temporal del celular correcto

Solo si ambos son válidos:

- permite acceso

¿Para qué sirve esto?

Hace el acceso mucho más seguro porque aunque alguien robe:

- tu contraseña
- o incluso tu llave SSH

todavía necesitaría:

- tu celular con Google Authenticator para poder entrar al servidor.

Paso 7. Verificación de errores

Agregar el comando:

```
sudo sshd -t
```

```
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$ sudo sshd -t  
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$
```

Si no da errores, todo está bien.

Sirve para verificar si la configuración del servidor SSH tiene errores antes de reiniciar el servicio.

Explicación:

- `sudo`

Ejecuta como administrador.

- `sshd`

Es el servidor SSH (daemon de SSH).

- `-t`

Significa “test”.

Hace una prueba de configuración.

¿Qué revisa?

Verifica archivos como:

`/etc/ssh/sshd_config`

y detecta:

- errores de sintaxis
- opciones mal escritas
- configuraciones incompatibles
- parámetros inválidos

Paso 8. Reiniciar SSH

```
sudo systemctl restart ssh
```

```
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$ sudo systemctl restart ssh  
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$ |
```

9. INSTALAR FAIL2BAN

¿Qué es?

Sistema que detecta:

- intentos fallidos
- ataques brute force

Y bloquea automáticamente la IP atacante.

Paso 1. Instalar

```
sudo apt install fail2ban -y
```

```
seguridad@eqnaranja:~$ sudo apt install fail2ban -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  mailx monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 496 kB of archives.
After this operation, 2572 kB of additional disk space will be used.
Get:1 http://mirrors.digitalocean.com/ubuntu noble/main amd64 python3-pyasn
core all 1.0.2-2 [10.1 kB]
Get:2 http://mirrors.digitalocean.com/ubuntu noble-updates/universe amd64 fa
il2ban all 1.0.2-3ubuntu0.1 [409 kB]
Get:3 http://mirrors.digitalocean.com/ubuntu noble/main amd64 python3-pyinot
ify all 0.9.6-2ubuntu1 [25.0 kB]
Get:4 http://mirrors.digitalocean.com/ubuntu noble/main amd64 whois amd64 5.
5.22 [51.7 kB]
Fetched 496 kB in 0s (6506 kB/s)
Selecting previously unselected package python3-pyasyncore.
(Reading database ... 110580 files and directories currently installed.)
Preparing to unpack .../python3-pyasyncore_1.0.2-2_all.deb ...
Unpacking python3-pyasyncore (1.0.2-2) ...
Selecting previously unselected package fail2ban.
Preparing to unpack .../fail2ban_1.0.2-3ubuntu0.1_all.deb ...
Unpacking fail2ban (1.0.2-3ubuntu0.1) ...
Selecting previously unselected package python3-pyinotify.
Preparing to unpack .../python3-pyinotify_0.9.6-2ubuntu1_all.deb ...
Unpacking python3-pyinotify (0.9.6-2ubuntu1) ...
Selecting previously unselected package whois.
Preparing to unpack .../whois_5.5.22_amd64.deb ...
Unpacking whois (5.5.22) ...
Setting up whois (5.5.22) ...
Setting up python3-pyasyncore (1.0.2-2) ...
```

Paso 2. Copiar configuración

```
sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
```

```
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$
```

¿Por qué?

Nunca se modifica jail.conf directamente.

Paso 3. Editar configuración

`sudo nano /etc/fail2ban/jail.local`

Paso 4. Configurar SSH

Buscar:

[sshd]

```
[sshd]
# To use more aggressive sshd modes set filter parameter "mode" in jail.local:
# normal (default), ddos, extra or aggressive (combines all).
# See "tests/files/logs/sshd" or "filter.d/sshd.conf" for usage example and de>
#mode    = normal
port     = ssh
logpath  = %(sshd_log)s
backend  = %(sshd_backend)s
```

Cambiar:

```
enabled = true
port = 1222
maxretry = 3
bantime = 3600
```

Explicación

Línea	Función
enabled	Activa protección
port	Puerto SSH
maxretry	Intentos permitidos
bantime	Tiempo de bloqueo

```
[sshd]

# To use more aggressive sshd modes set filter parameter "mode" in jail.local:
# normal (default), ddos, extra or aggressive (combines all).
# See "tests/files/logs/sshd" or "filter.d/sshd.conf" for usage example and de>
#mode    = normal
#port    = ssh
#logpath = %(sshd_log)s
#backend = %(sshd_backend)s

enabled = true

port = 1222

maxretry = 3

bantime = 3600
```

Paso 5. Reiniciar servicio

`sudo systemctl restart fail2ban`

```
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo systemctl restart fail2ban
seguridad@eqnaranja:~$ |
```

10. HARDENING MARIADB

Objetivo

Proteger base de datos.

Paso 1. Instalar MariaDB

`sudo apt install mariadb-server -y`

```

seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo apt install mariadb-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  galera-4 libcgi-fast-perl libcgi-pm-perl libencode-locale-perl libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl
  libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libsnappy1v5 libtimedate-perl
  liburi-perl liburing2 mariadb-plugin-provider-bzip2
  mariadb-plugin-provider-lz4 mariadb-plugin-provider-lzma
  mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy
  mariadb-server-core pv socat
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl
  libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx mariadb-test
  doc-base
The following NEW packages will be installed:
  galera-4 libcgi-fast-perl libcgi-pm-perl libencode-locale-perl libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl
  libhtml-template-perl libhttp-date-perl libhttp-message-perl
  libio-html-perl liblwp-mediatypes-perl libsnappy1v5 libtimedate-perl
  liburi-perl liburing2 mariadb-plugin-provider-bzip2
  mariadb-plugin-provider-lz4 mariadb-plugin-provider-lzma
  mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy mariadb-server
  mariadb-server-core pv socat
0 upgraded, 27 newly installed, 0 to remove and 0 not upgraded.
Need to get 13.8 MB of archives.
After this operation, 110 MB of additional disk space will be used.
Get:1 http://mirrors.digitalocean.com/ubuntu noble/universe amd64 galera-4 amd64 26.4.16-2build4 [736 kB]
Get:2 http://mirrors.digitalocean.com/ubuntu noble/main amd64 liburing2 amd64 2.5-1build1 [21.1 kB]
Get:3 http://mirrors.digitalocean.com/ubuntu noble-updates/universe amd64 mariadb-server-core amd64 1:10.11.14-0ubuntu0.24.04.1 [8406 kB]
Get:4 http://mirrors.digitalocean.com/ubuntu noble/main amd64 socat amd64 1.8.0-4build3 [374 kB]
Get:5 http://mirrors.digitalocean.com/ubuntu noble-updates/universe amd64 maria

```

Paso 2. Ejecutar asistente

```
sudo mysql_secure_installation
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
seguridad@eqnaranja:~$ sudo mysql_secure_installation
```

```
NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB  
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!
```

```
In order to log into MariaDB to secure it, we'll need the current  
password for the root user. If you've just installed MariaDB, and  
haven't set the root password yet, you should just press enter here.
```

```
Enter current password for root (enter for none):  
OK, successfully used password, moving on...
```

```
Setting the root password or using the unix_socket ensures that nobody  
can log into the MariaDB root user without the proper authorisation.
```

```
You already have your root account protected, so you can safely answer 'n'.
```

```
Switch to unix_socket authentication [Y/n] y  
Enabled successfully!  
Reloading privilege tables..  
... Success!
```

```
You already have your root account protected, so you can safely answer 'n'.
```

```
Change the root password? [Y/n] y  
New password:  
Re-enter new password:  
Password updated successfully!  
Reloading privilege tables..  
... Success!
```

```
By default, a MariaDB installation has an anonymous user, allowing anyone  
to log into MariaDB without having to have a user account created for  
them. This is intended only for testing, and to make the installation  
go a bit smoother. You should remove them before moving into a
```

¿Qué hace?

Permite:

- poner contraseña root
- borrar usuarios anónimos
- eliminar base de pruebas
- bloquear root remoto

11. HARDENING POSTGRESQL

Paso 1. Instalar PostgreSQL

`sudo apt install postgresql postgresql-contrib -y`

```
seguridad@eqnaranja:~$  
seguridad@eqnaranja:~$ sudo apt install postgresql postgresql-contrib -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm17t64 libpq5  
  libtypes-serialiser-perl postgresql-16 postgresql-client-16  
  postgresql-client-common postgresql-common ssl-cert  
Suggested packages:  
  postgresql-doc postgresql-doc-16  
The following NEW packages will be installed:  
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm17t64 libpq5  
  libtypes-serialiser-perl postgresql postgresql-16 postgresql-client-16  
  postgresql-client-common postgresql-common postgresql-contrib ssl-cert  
0 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.  
Need to get 43.6 MB of archives.  
After this operation, 175 MB of additional disk space will be used.  
Get:1 http://mirrors.digitalocean.com/ubuntu noble/main amd64 libjson-perl all 4.10000-1 [81.9 kB]  
Get:2 http://mirrors.digitalocean.com/ubuntu noble-updates/main amd64 postgresql-client-common all 257build1.1 [36.4 kB]  
Get:3 http://mirrors.digitalocean.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]  
Get:4 http://mirrors.digitalocean.com/ubuntu noble-updates/main amd64 postgresql-common all 257build1.1 [161 kB]  
Get:5 http://mirrors.digitalocean.com/ubuntu noble/main amd64 libcommon-sense-perl amd64 3.75-3build3 [20.4 kB]  
Get:6 http://mirrors.digitalocean.com/ubuntu noble/main amd64 libtypes-serialiser-perl all 1.01-1 [11.6 kB]  
Get:7 http://mirrors.digitalocean.com/ubuntu noble-updates/main amd64 libjson-xs-perl amd64 4.040-0ubuntu0.24.04.1 [83.7 kB]  
Get:8 http://mirrors.digitalocean.com/ubuntu noble/main amd64 libllvm17t64 amd64 1:17.0.6-9ubuntu1 [26.2 MB]  
Get:9 http://mirrors.digitalocean.com/ubuntu noble-updates/main amd64 libpq5 amd64 16.14-0ubuntu0.24.04.1 [147 kB]  
Get:10 http://mirrors.digitalocean.com/ubuntu noble-updates/main amd64 postgresql-client-16 amd64 16.14-0ubuntu0.24.04.1 [1300 kB]
```

Paso 2. Cambiar contraseña

`sudo passwd postgres`

```
seguridad@eqnaranja:~$ sudo passwd postgres
New password:
Retype new password:
passwd: password updated successfully
seguridad@eqnaranja:~$ |
```

Paso 3. Entrar

sudo -u postgres psql

```
passwd: password updated successfully
seguridad@eqnaranja:~$ sudo -u postgres psql
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

postgres=#
```

Paso 4. Cambiar contraseña interna

ALTER USER postgres PASSWORD 'Segura#2026';

```
seguridad@eqnaranja:~$ sudo -u postgres psql
psql (16.14 (Ubuntu 16.14-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# ALTER USER postgres PASSWORD 'EquiNaranja123';
ALTER ROLE
postgres=# |
```

12. VERIFICACIÓN

Parte 1.

Root no tiene acceso

```
PS C:\Users\maryp> ssh root@159.203.33.12 -p 1222
root@159.203.33.12: Permission denied (publickey).
PS C:\Users\maryp> |
```

Parte 2.

PC Local puede entrar con contraseña y código de verificación, teniendo en cuenta que ya esta automáticamente las llaves anteriormente

```
PS C:\Users\maryp> ssh seguridad@159.203.33.12 -p 1222
(seguridad@159.203.33.12) Password:
(seguridad@159.203.33.12) Verification code:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu May 28 19:03:06 UTC 2026

System load:          0.0
Usage of /:           13.6% of 23.17GB
Memory usage:         38%
Swap usage:           0%
Processes:            134
Users logged in:      2
IPv4 address for eth0: 159.203.33.12
IPv4 address for eth0: 10.20.0.5
IPv6 address for eth0: 2604:a880:cad:d0:0:1:83f5:1001

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM service at https://ubuntu.com/esm

*** System restart required ***
Last login: Thu May 28 18:07:13 2026 from 187.194.208.110
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ |
```

Parte 3.

Accesoo denegado a PC sin llaves de acceso

```
PS C:\Users\karla> ssh seguridad@159.203.33.12 -p 1222
seguridad@159.203.33.12: Permission denied (publickey).
```

Parte 4.

Ver firewall:

sudo ufw status verbose

```
seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo ufw status verbose
[sudo] password for seguridad:
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
2222/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
22/tcp DENY IN Anywhere
1222/tcp ALLOW IN Anywhere
2222/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
443/tcp (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) DENY IN Anywhere (v6)
1222/tcp (v6) ALLOW IN Anywhere (v6)

seguridad@eqnaranja:~$ |
```

Ver usuarios:

cat /etc/passwd

```

seguridad@eqnaranja:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
tss:x:104:104:TPM software stack,,,:/var/lib/tpm:/bin/false
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
pollinate:x:106:1::/var/cache/pollinate:/bin/false
tcpdump:x:107:108::/nonexistent:/usr/sbin/nologin
landscape:x:108:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:990:990:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
polkitd:x:989:989:User for polkitd:/:/usr/sbin/nologin
mysql:x:109:112:MySQL Server,,,:/nonexistent:/bin/false
seguridad:x:1000:1000:naranja,eqnaranja,554944065,4272247835:/home/seguridad:/bin/bash
_galera:x:110:65534::/nonexistent:/usr/sbin/nologin
postgres:x:111:114:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
seguridad@eqnaranja:~$ |

```

Ver logs SSH:

sudo journalctl -u ssh

```

seguridad@eqnaranja:~$
seguridad@eqnaranja:~$ sudo journalctl -u ssh
May 28 02:56:49 eqnaranja systemd[1]: Starting ssh.service - OpenBSD Secure Sh>
May 28 02:56:49 eqnaranja sshd[1828]: Server listening on 0.0.0.0 port 22.
May 28 02:56:49 eqnaranja sshd[1828]: Server listening on :: port 22.
May 28 02:56:49 eqnaranja systemd[1]: Started ssh.service - OpenBSD Secure She>
May 28 02:57:23 eqnaranja sshd[1830]: Accepted password for root from 131.196.>
May 28 02:57:23 eqnaranja sshd[1830]: pam_unix(sshd:session): session opened f>
May 28 03:05:56 eqnaranja sshd[2003]: Accepted password for root from 189.180.>
May 28 03:05:56 eqnaranja sshd[2003]: pam_unix(sshd:session): session opened f>
May 28 03:06:38 eqnaranja sshd[2080]: Connection closed by 47.107.112.39 port >
May 28 03:22:14 eqnaranja sshd[2109]: Connection closed by 194.59.206.2 port 3>
May 28 03:26:44 eqnaranja sshd[2115]: pam_unix(sshd:auth): authentication fail>
May 28 03:26:46 eqnaranja sshd[2115]: Failed password for root from 43.139.39.>
May 28 03:38:18 eqnaranja sshd[2199]: Accepted password for root from 189.180.>
May 28 03:38:18 eqnaranja sshd[2199]: pam_unix(sshd:session): session opened f>
May 28 04:08:02 eqnaranja systemd[1]: Stopping ssh.service - OpenBSD Secure Sh>
May 28 04:08:02 eqnaranja sshd[1828]: Received signal 15; terminating.
May 28 04:08:02 eqnaranja systemd[1]: ssh.service: Deactivated successfully.
May 28 04:08:02 eqnaranja systemd[1]: Stopped ssh.service - OpenBSD Secure She>
May 28 04:22:10 eqnaranja systemd[1]: Starting ssh.service - OpenBSD Secure Sh>
May 28 04:22:10 eqnaranja sshd[20324]: Server listening on 0.0.0.0 port 22.
May 28 04:22:10 eqnaranja sshd[20324]: Server listening on :: port 22.
May 28 04:22:10 eqnaranja systemd[1]: Started ssh.service - OpenBSD Secure She>
May 28 04:22:11 eqnaranja sshd[20325]: Connection closed by 194.59.206.2 port >
May 28 04:31:54 eqnaranja sshd[20388]: Invalid user eqnaranja from 189.180.1.1>
May 28 04:32:01 eqnaranja sshd[20388]: pam_unix(sshd:auth): check pass; user u>
May 28 04:32:01 eqnaranja sshd[20388]: pam_unix(sshd:auth): authentication fai>
May 28 04:32:03 eqnaranja sshd[20388]: Failed password for invalid user eqnara>
May 28 04:32:14 eqnaranja sshd[20388]: pam_unix(sshd:auth): check pass; user u>
May 28 04:32:16 eqnaranja sshd[20388]: Failed password for invalid user eqnara>
May 28 04:33:30 eqnaranja sshd[20390]: Invalid user eqnaranja from 189.180.1.1>
May 28 04:33:34 eqnaranja sshd[20388]: Connection reset by invalid user eqnara>
May 28 04:33:34 eqnaranja sshd[20388]: PAM 1 more authentication failure; logn>
May 28 04:33:38 eqnaranja sshd[20390]: pam_unix(sshd:auth): check pass; user u>
May 28 04:33:38 eqnaranja sshd[20390]: pam_unix(sshd:auth): authentication fai>
May 28 04:33:40 eqnaranja sshd[20390]: Failed password for invalid user eqnara>
May 28 04:33:51 eqnaranja sshd[20390]: pam_unix(sshd:auth): check pass; user u>
May 28 04:33:53 eqnaranja sshd[20390]: Failed password for invalid user eqnara>
May 28 04:34:14 eqnaranja sshd[20392]: Invalid user eqnaranja from 189.180.1.1>
May 28 04:34:17 eqnaranja sshd[20390]: Connection reset by invalid user eqnara>
May 28 04:34:17 eqnaranja sshd[20390]: PAM 1 more authentication failure; logn>
lines 1-40...skipping...
May 28 02:56:49 eqnaranja systemd[1]: Starting ssh.service - OpenBSD Secure Shell ser>
May 28 02:56:49 eqnaranja sshd[1828]: Server listening on 0.0.0.0 port 22.
May 28 02:56:49 eqnaranja sshd[1828]: Server listening on :: port 22.
May 28 02:56:49 eqnaranja systemd[1]: Started ssh.service - OpenBSD Secure Shell serv>
May 28 02:57:23 eqnaranja sshd[1830]: Accepted password for root from 131.196.245.181>

```

13. RESPALDOS AUTOMÁTICOS 3-2-1

¿Por qué son importantes?

Los respaldos son fundamentales dentro del hardening de un servidor, ya que permiten recuperar información en caso de:

- ransomware
- borrado accidental

- fallos de disco
- corrupción de datos
- ataques informáticos

La estrategia 3-2-1 consiste en:

Regla	Significado
3 copias	Tener 3 copias de la información
2 medios distintos	Guardar respaldos en diferentes ubicaciones
1 copia externa	Mantener una copia fuera del servidor

PASO 1. Crear estructura de respaldos (3-2-1)

En este paso se crean las carpetas donde se almacenarán los distintos niveles de respaldo.

```
mkdir -p ~/api/backups/local
mkdir -p /var/backups/api
mkdir -p /var/backups/api/temp
```

Explicación del comando

Parte	Función
mkdir	Crea directorios
-p	Crea rutas completas automáticamente
~/api/backups/local	Respaldo local dentro de la aplicación
/var/backups/api	Respaldo secundario del sistema
/var/backups/api/temp	Archivos temporales antes de comprimir

RESUMEN SIMPLE

Ruta	Función
/var/backups/api	Segundo respaldo seguro y externo
/var/backups/api/temp	Archivos temporales antes de comprimir
~/api/backups/local	Primera copia local de seguridad
sudo	Permisos administrativos

```
seguridad@eqnaranja:~$ mkdir -p ~/api/backups/local
seguridad@eqnaranja:~$ sudo mkdir -p /var/backups/api
[sudo] password for seguridad:
seguridad@eqnaranja:~$ sudo mkdir -p /var/backups/api/temp
```

```
seguridad@eqnaranja:~$ sudo chmod -R 777 /var/backups/api
```

Explicación COMPLETA

Parte	Función
chmod	Cambia permisos
-R	Aplica recursivamente
777	Permisos totales
/var/backups/api	Carpeta afectada

¿Qué significa 777?

Número Permisos

7 lectura + escritura + ejecución

Entonces:

777 = TODOS PUEDEN HACER TODO

PASO 2. Identificar datos importantes de la API

La API contiene información crítica que debe protegerse mediante respaldos automáticos.

Entre los datos importantes se encuentran:

- fotografías de usuarios
- archivos subidos

- base de datos SQL
- configuraciones del sistema

```
seguridad@eqnaranja:~$ cd api
seguridad@eqnaranja:~/api$ ls
'Base de Datos.dia'  Instalar.md  README.md  backups  db.sql  index.php  localhost+1-key.pem
'Base de Datos.png'  LICENSE      api.log    db.json  htaccess.txt  index.py    localhost+1.pem
```

PASO 3. Crear script de respaldo

Se crea un script automatizado que realizará los respaldos de manera automática.

nano respaldo.sh

¿Qué hace este comando?

Parte	Función
nano	Editor de texto
respaldo.sh	Archivo script Bash

```
seguridad@eqnaranja:~$ nano respaldo.sh
```

PASO 4. Agregar script de respaldo (3-2-1 COMPLETO)

Dentro del archivo se agrega el siguiente script:

```
#!/bin/bash
```

```
FECHA=$(date +"%Y-%m-%d_%H-%M-%S")
```

```
# RUTAS
```

```
BACKUP_LOCAL="$HOME/api/backups/local"
```

```
BACKUP_SECUNDARIO="/var/backups/api"
```

```
TEMP="/var/backups/api/temp"
```

```
# BASE DE DATOS (USUARIO SEGURO)
```

```
DB_NAME="basedatos"
```

```
DB_USER="eqnaranja"
```

```
DB_PASS="EquiNaranja123e"
```

```
RUTA_FOTOS="$HOME/api "
```

```
mkdir -p $BACKUP_LOCAL
mkdir -p $BACKUP_SECUNDARIO
mkdir -p $TEMP
```

```
echo "Iniciando respaldo 3-2-1..."
```

```
# =====
```

```
# 1) RESPALDO SQL
```

```
# =====
```

```
mysqldump -u $DB_USER -p$DB_PASS $DB_NAME > $TEMP/db_$FECHA.sql
```

```
# =====
```

```
# 2) RESPALDO FOTOS
```

```
# =====
```

```
cp -r $RUTA_FOTOS $TEMP/
```

```
# =====
```

```
# 3) PRIMER RESPALDO (LOCAL)
```

```
# =====
```

```
cp -r $TEMP/* $BACKUP_LOCAL/
```

```
# =====
```

```
# 4) SEGUNDO RESPALDO (TAR)
```

```
# =====
```

```
tar -czvf $BACKUP_SECUNDARIO/respaldo_$FECHA.tar.gz -C $TEMP .
```

```
# =====
```

```
# 5) TERCER RESPALDO (NUBE)
```

```
# =====
```

```
rclone copy $BACKUP_SECUNDARIO/respaldo_$FECHA.tar.gz drive:RespalDOSAPI
```

```
# LIMPIEZA
```

```
rm -rf $TEMP/*
```

```
echo "Respaldo completado"
```

Explicación COMPLETA del script

Parte	Función
mysqldump	Genera respaldo de la base de datos SQL

cp -r	Copia carpetas y archivos recursivamente
uploads	Carpeta que contiene imágenes y archivos
local backup	Primera copia local dentro del servidor
tar	Comprime archivos en formato .tar.gz
/var/backups/api	Segundo respaldo fuera de la aplicación
yclone	Envía respaldo a almacenamiento en nube
temp	Carpeta temporal para evitar duplicación

```
seguridad@eqnaranja:~$ sudo mysql -u eqnaranja -pEquiNaranja123e
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 49
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| basedatos |
| information_schema |
+-----+
2 rows in set (0.000 sec)

MariaDB [(none)]> exit;
Bye
```

```
seguridad@eqnaranja: ~  
GNU nano 7.2 respaldo.sh  
#!/bin/bash  
  
FECHA=$(date +%Y-%m-%d_%H-%M-%S")  
  
# RUTAS  
BACKUP_LOCAL="$HOME/api/backups/local"  
BACKUP_SECUNDARIO="/var/backups/api"  
TEMP="/var/backups/api/temp"  
  
# BASE DE DATOS (USUARIO SEGURO)  
DB_NAME="basedatos"  
DB_USER="eqnaranja"  
DB_PASS="EquiNaranja123e"  
  
RUTA_FOTOS="$HOME/api"  
  
mkdir -p $BACKUP_LOCAL  
mkdir -p $BACKUP_SECUNDARIO  
mkdir -p $TEMP  
  
echo "Iniciando respaldo 3-2-1..."  
  
# =====  
# 1) RESPALDO SQL  
# =====  
mysqldump -u $DB_USER -p$DB_PASS $DB_NAME > $TEMP/db_$FECHA.sql  
[ Read 51 lines ]  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo      M-6 Copy  
  
seguridad@eqnaranja: ~  
GNU nano 7.2 respaldo.sh *  
# =====  
# 2) RESPALDO FOTOS  
# =====  
cp -r $RUTA_FOTOS $TEMP/  
  
# =====  
# 3) PRIMER RESPALDO (LOCAL)  
# =====  
cp -r $TEMP/* $BACKUP_LOCAL/  
  
# =====  
# 4) SEGUNDO RESPALDO (TAR)  
# =====  
tar -czvf $BACKUP_SECUNDARIO/respaldo_$FECHA.tar.gz -C $TEMP .  
  
# =====  
# 5) TERCER RESPALDO (NUBE)  
# =====  
rclone copy $BACKUP_SECUNDARIO/respaldo_$FECHA.tar.gz drive:RespalDOSAPI  
  
# LIMPIEZA  
rm -rf $TEMP/*  
  
echo "Respaldo completado"
```

¿Qué hace exactamente este sistema?

El script:

1. genera un respaldo de la base de datos
2. copia las imágenes de la API
3. crea un respaldo local

4. genera un archivo comprimido
5. sube una copia a la nube
6. limpia archivos temporales

Esto implementa correctamente la estrategia de respaldo 3-2-1.

PASO 5. Dar permisos de ejecución

```
chmod +x respaldo.sh
```

Explicación

Parte	Función
chmod	Cambia permisos
+x	Permite ejecutar
respaldo.sh	Script de respaldo

¿Qué hace?

Convierte el archivo en un script ejecutable.

```
seguridad@eqnaranja:~$ chmod +x respaldo.sh
```

PROBAR MANUALMENTE

```
seguridad@eqnaranja: ~  
seguridad@eqnaranja:~$ bash respaldo.sh  
Iniciando respaldo 3-2-1...  
./  
./api/  
./api/localhost+1-key.pem  
./api/index.php  
./api/README.md  
./api/backups/  
./api/backups/local/  
./api/backups/local/db_2026-05-28_21-14-11.sql  
./api/backups/local/db_2026-05-28_21-17-29.sql  
./api/db.sql  
./api/Base de Datos.png  
./api/db.json  
./api/Base de Datos.dia  
./api/localhost+1.pem  
./api/Instalar.md  
./api/.git/  
./api/.git/branches/  
./api/.git/description  
./api/.git/packed-refs  
./api/.git/index  
./api/.git/HEAD  
./api/.git/objects/  
./api/.git/objects/info/  
./api/.git/objects/pack/  
./api/.git/objects/pack/pack-d31069884f22a17ccdfaaa6d600dc959f9742eba.rev  
./api/.git/objects/pack/pack-d31069884f22a17ccdfaaa6d600dc959f9742eba.pack  
./api/.git/objects/pack/pack-d31069884f22a17ccdfaaa6d600dc959f9742eba.idx  
./api/.git/info/  
./api/.git/info/exclude  
./api/.git/logs/  
./api/.git/logs/HEAD  
./api/.git/logs/refs/  
./api/.git/logs/refs/remotes/  
./api/.git/logs/refs/remotes/origin/  
./api/.git/logs/refs/remotes/origin/HEAD  
./api/.git/logs/refs/heads/  
./api/.git/logs/refs/heads/main  
./api/.git/refs/  
./api/.git/refs/remotes/
```

VERIFICAR PRIMER RESPALDO

```
seguridad@eqnaranja:~$ ls ~/api/backups/local  
api db_2026-05-28_21-14-11.sql db_2026-05-28_21-17-29.sql db_2026-05-28_21-19-30.sql
```

SEGUNDO RESPALDO

```
seguridad@eqnaranja:~$ ls /var/backups/api  
respaldo_2026-05-28_21-14-11.tar.gz  respaldo_2026-05-28_21-17-29.tar.gz  respaldo_2026-05-28_21-19-30.tar.gz
```

PASO 6 — CONFIGURAR TERCER RESPALDO EN LA NUBE

La tercera copia de seguridad corresponde al respaldo externo en la nube, cumpliendo la estrategia 3-2-1.

Las plataformas más utilizadas son:

- Google Drive
- Dropbox
- Otro VPS
- OneDrive

Para servidores Linux se utilizará la herramienta:

rclone

¿Qué es rclone?

rclone es una herramienta que permite sincronizar archivos entre Linux y servicios de almacenamiento en nube.

Permite:

- subir respaldos automáticamente
- sincronizar archivos
- automatizar copias remotas
- proteger información fuera del servidor

PASO 1. Instalar rclone

```
sudo apt install rclone -y
```

Explicación

Parte Función

apt install Instala paquetes

rclone Herramienta de sincronización nube

-y Acepta automáticamente

```
seguridad@eqnaranja:~$ sudo apt update
sudo apt install rclone -y
Hit:1 http://mirrors.digitalocean.com/ubuntu noble InRelease
Hit:2 https://repos-droplet.digitalocean.com/apt/droplet-agent main InRelease
Hit:3 http://mirrors.digitalocean.com/ubuntu noble-updates InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://mirrors.digitalocean.com/ubuntu noble-backports InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1704 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [42.5 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1192 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Fetched 3139 kB in 12s (270 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  rclone
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 16.5 MB of archives.
After this operation, 62.6 MB of additional disk space will be used.
Get:1 http://mirrors.digitalocean.com/ubuntu noble-updates/universe amd64 rclone amd64 1.60.1+dfsg-3ubuntu0.24.04.5 [16.5 MB]
Fetched 16.5 MB in 0s (44.6 MB/s)
Selecting previously unselected package rclone.
(Reading database ... 113661 files and directories currently installed.)
Preparing to unpack .../rclone_1.60.1+dfsg-3ubuntu0.24.04.5_amd64.deb ...
Unpacking rclone (1.60.1+dfsg-3ubuntu0.24.04.5) ...
Setting up rclone (1.60.1+dfsg-3ubuntu0.24.04.5) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-71-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-117-generic.
```

PASO 2. Configurar almacenamiento nube

Ejecutar:

rclone config

```

seguridad@eqnaranja:~$ rclone config
2026/05/28 21:37:16 NOTICE: Config file "/home/seguridad/.config/rclone/rclone.conf" not found - using defaults
No remotes found, make a new one?
n) New remote
s) Set configuration password
q) Quit config
n/s/q> n

Enter name for new remote.
name> drivebackup

Option Storage.
Type of storage to configure.
Choose a number from below, or type in your own value.
 1 / 1Fichier
   \ (fichier)
 2 / Akamai NetStorage
   \ (netstorage)
 3 / Alias for an existing remote
   \ (alias)
 4 / Amazon Drive
   \ (amazon cloud drive)
 5 / Amazon S3 Compliant Storage Providers including AWS, Alibaba, Ceph, China Mobile, Cloudflare, ArvanCloud, Digital Ocean, Dreamhost, Huawei OBS, IBM COS
   \ (s3)
 6 / Backblaze B2
   \ (b2)
 7 / Better checksums for other remotes
   \ (hasher)
 8 / Box
   \ (box)
 9 / Cache a remote
   \ (cache)
10 / Citrix Sharefile
   \ (sharefile)
11 / Combine several remotes into one
   \ (combine)
12 / Compress a remote
   \ (compress)
13 / Dropbox
   \ (dropbox)

```

¿Qué hace?

Inicia el asistente de configuración de almacenamiento remoto.

Seleccionar proveedor

Elegir:

Google Drive

```

18 / Google Drive
   \ (drive)
19 / Google Photos

```

CONFIGURACIÓN DESDE WINDOWS (GENERAR TOKEN)

En Windows entrar a la carpeta:

```
cd .\rclone-v1.74.1-windows-amd64\
```

PASO 3. Verificar rclone.exe

Ejecutar:

```
ls
```

Debe aparecer:

rclone.exe

PASO 4. Verificar funcionamiento

.\rclone.exe version

Resultado esperado

rclone v1.74.1

PASO 5. Generar token de autenticación

Ejecutar:

.\rclone.exe authorize drive

```
seguridad@eqnaranja: ~
Windows PowerShell
PS C:\Users\karla> cd "C:\Users\karla\Downloads\rclone-v1.74.2-windows-amd64"
PS C:\Users\karla\Downloads\rclone-v1.74.2-windows-amd64> ls

Directorio: C:\Users\karla\Downloads\rclone-v1.74.2-windows-amd64

Mode                LastWriteTime         Length Name
----                -
-a-----      28/05/2026   03:45 p. m.         11146 git-log.txt
-a-----      28/05/2026   03:45 p. m.        3252473 rclone.1
-a-----      28/05/2026   03:45 p. m.        78708736 rclone.exe
-a-----      28/05/2026   03:45 p. m.        3740438 README.html
-a-----      28/05/2026   03:45 p. m.        2935779 README.txt

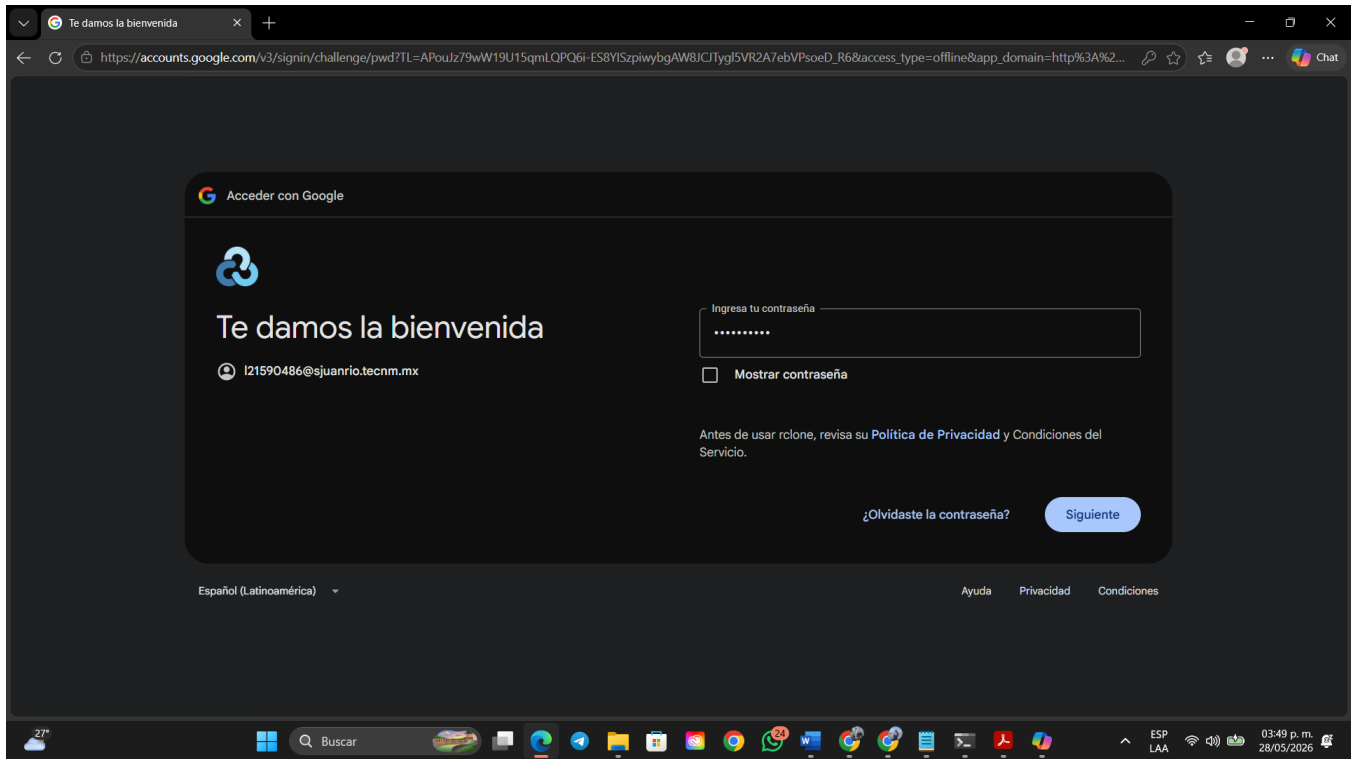
PS C:\Users\karla\Downloads\rclone-v1.74.2-windows-amd64> .\rclone.exe version
rclone v1.74.2
- os/version: Microsoft Windows 11 Home Single Language 25H2 25H2 (64 bit)
- os/kernel: 10.0.26290.8457 (x86_64)
- os/type: windows
- os/arch: amd64
- go/version: go1.26.3
- go/linking: static
- go/tags: cmount
PS C:\Users\karla\Downloads\rclone-v1.74.2-windows-amd64> .\rclone.exe authorize "drive" "eyJzY29wZSI6ImRyaXZlIn0"
2026/05/28 15:47:39 NOTICE: Config file "C:\Users\karla\AppData\Roaming\rclone\rclone.conf" not found - using defaults
2026/05/28 15:47:39 NOTICE: Make sure your Redirect URL is set to "http://127.0.0.1:53682/" in your custom config.
2026/05/28 15:47:39 NOTICE: If your browser doesn't open automatically go to the following link: http://127.0.0.1:53682/auth?state=uHYnePRW0DmPL66iADvUNQ
2026/05/28 15:47:39 NOTICE: Log in and authorize rclone for access
2026/05/28 15:47:39 NOTICE: Waiting for code...
```

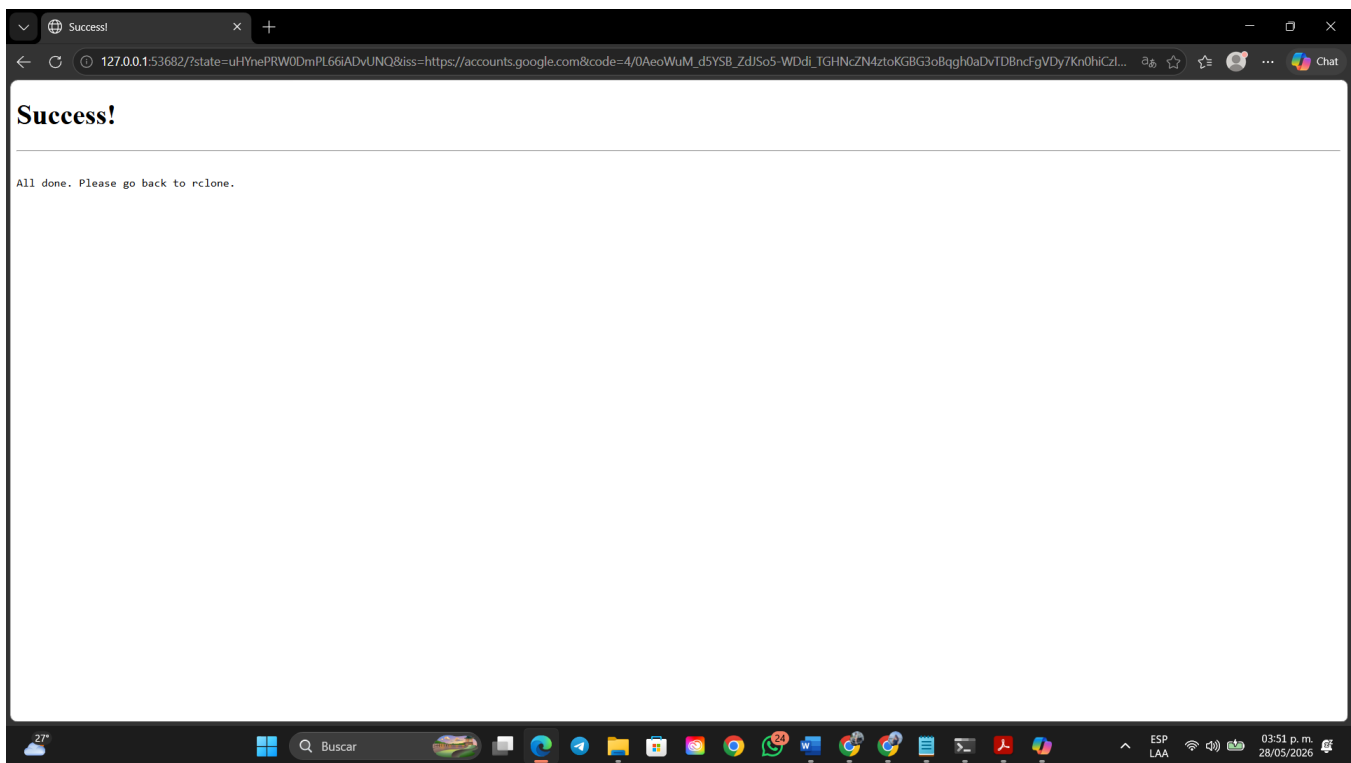
¿Qué sucede?

1. Se abre el navegador
2. Se inicia sesión en Google Drive
3. Se aceptan permisos
4. Google genera un token de autenticación

Ejemplo:

```
{"access_token":"...","refresh_token":"..."}
```





La configuración de autenticación de Google Drive tuvo que realizarse desde Windows debido a que el servidor Ubuntu funciona como una máquina headless, es decir, sin entorno gráfico ni navegador web. Google Drive utiliza autenticación OAuth, un sistema que requiere abrir un navegador para iniciar sesión y aceptar permisos de acceso.

Como el VPS no puede mostrar ventanas ni abrir Google Chrome o Firefox, fue necesario utilizar una computadora con interfaz gráfica, en este caso Windows, para generar el token de autenticación. Desde Windows se ejecutó `rclone.exe authorize drive`, lo que permitió abrir el navegador, iniciar sesión en Google y autorizar el acceso a la cuenta de Google Drive.

Una vez completada la autenticación, Google generó un token seguro con los permisos necesarios. Ese token posteriormente se copió y se pegó en el servidor Ubuntu durante la configuración de rclone. Gracias a este procedimiento, el servidor puede conectarse y subir respaldos automáticamente a Google Drive sin necesidad de contar con navegador o entorno gráfico instalado.

PASO 6. Volver al servidor Linux

En Linux ejecutar:

`rclone config reconnect drivebackup:`

o cuando aparezca:

config_token>

Pegar el JSON copiado anteriormente.

RESULTADO

El servidor quedará conectado con Google Drive.

```
seguridad@eqnaranja:~$ rclone listremotes
drivebackup:
seguridad@eqnaranja:~$ |
```

PASO 7 — PROBAR SUBIDA A LA NUBE

Ejecutar:

```
rclone copy /var/backups/api drivebackup:RespaldosAPI
```

```
seguridad@eqnaranja:~$ rclone copy /var/backups/api drivebackup:RespaldosAPI
seguridad@eqnaranja:~$ rclone ls drivebackup:RespaldosAPI
100947 respaldo_2026-05-28_21-19-30.tar.gz
647 respaldo_2026-05-28_21-17-29.tar.gz
562 respaldo_2026-05-28_21-14-11.tar.gz
```

Explicación

Parte	Función
rclone copy	Copia archivos
/var/backups/api	Carpeta local
drivebackup:RespaldosAPI	Carpeta remota en Google Drive

¿Qué hace?

Sube automáticamente los respaldos generados hacia Google Drive.

Comprobación de que efectivamente están los respaldos en Google Drive

Carpeta de RespaldosAPI

Nombre	Propietario	Fecha de modificación	Tamaño de a	Ordenar
respaldo_2026-05-28_21-19-30.tar.gz	yo	15:19 yo	99 kB	
respaldo_2026-05-28_21-17-29.tar.gz	yo	15:17 yo	647 bytes	
respaldo_2026-05-28_21-14-11.tar.gz	yo	15:14 yo	562 bytes	

Contenido dentro de los respaldos

```
BTxt
[{"id": "1", "name": "Bot.txt", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "2", "name": "Terminal.txt", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "3", "name": "mysql-u root", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "4", "name": "reporte_limpieza.txt", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "5", "name": "ssh seguridad", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "6", "name": "localhost+1-key.pem", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "7", "name": "localhost+1.pem", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 100, "mime": "text/plain"}, {"id": "8", "name": "respaldo_2026-05-28_21-14-11.tar.gz", "type": "file", "parent": "1", "created": "2026-05-28T21:14:11Z", "modified": "2026-05-28T21:14:11Z", "size": 562, "mime": "application/tar-gz"}]
```

PASO 8 — AUTOMATIZAR RESPALDOS CON CRON

Abrir el programador de tareas:

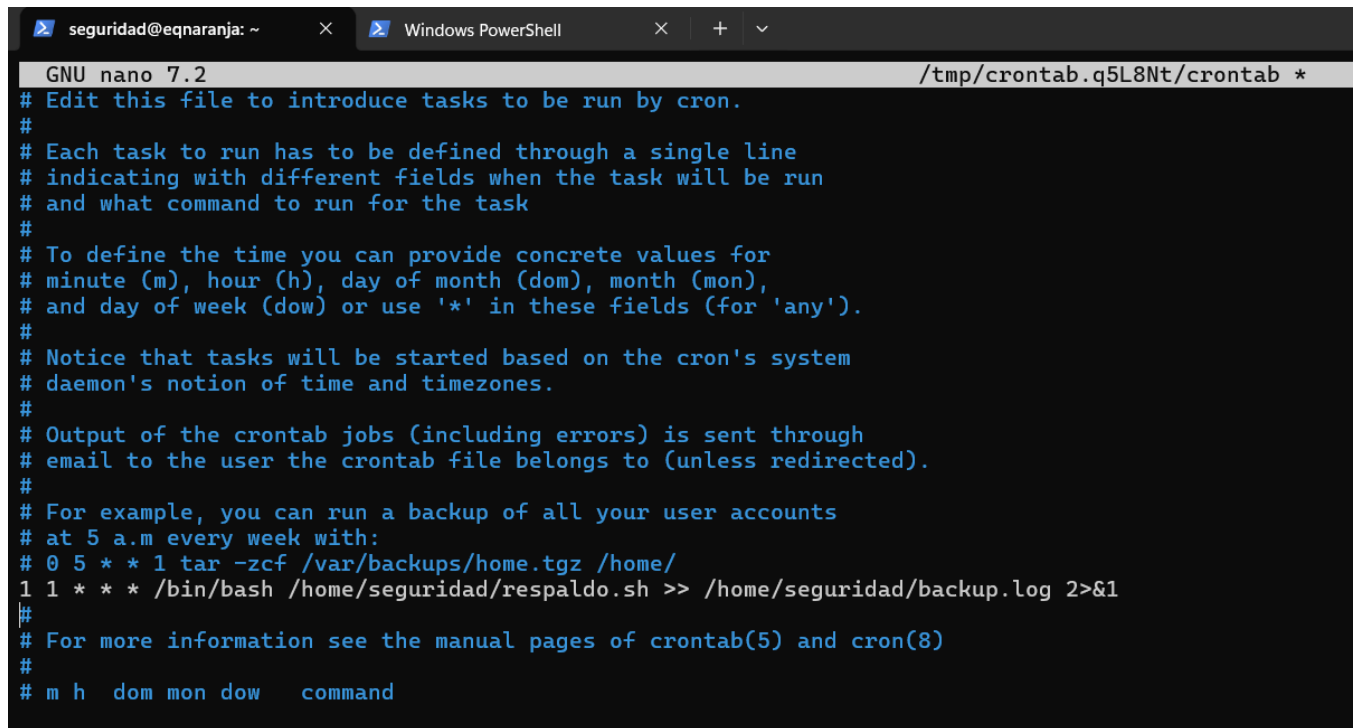
crontab

-e

```
seguridad@eqnaranja:~$ crontab -e
```

Agregar:

```
1 1 * * * /bin/bash /home/seguridad/respaldo.sh >> /home/seguridad/backup.log 2>&1
```



```
GNU nano 7.2 /tmp/crontab.q5L8Nt/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
1 1 * * * /bin/bash /home/seguridad/respaldo.sh >> /home/seguridad/backup.log 2>&1
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
```

¿Qué hace esta tarea?

Todos los días:

- 1:01 AM
- genera respaldo
- comprime archivos
- sube respaldo a nube
- guarda logs automáticos

Explicación del CRON

Campo Función

1 minuto
1 hora
* cualquier día
* cualquier mes
* cualquier semana

```
seguridad@eqnaranja:~$ systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-28 04:10:18 UTC; 17h ago
     Docs: man:cron(8)
  Main PID: 19352 (cron)
    Tasks: 1 (limit: 1110)
   Memory: 2.2M (peak: 9.2M)
      CPU: 5.699s
   CGroup: /system.slice/cron.service
           └─19352 /usr/sbin/cron -f -P

Warning: some journal files were not opened due to insufficient permissions.
seguridad@eqnaranja:~$ |
```

14. VERIFICACIONES FINALES

Ver puertos abiertos

sudo ss -tulnp

```
seguridad@eqnaranja:~$ sudo ss -tulnp
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
udp    UNCONN 0        0       127.0.0.54:53        0.0.0.0:*          users:(("systemd-resolve",pid=9942,fd=16))
udp    UNCONN 0        0       127.0.0.53%lo:53     0.0.0.0:*          users:(("systemd-resolve",pid=9942,fd=14))
tcp    LISTEN 0        80       127.0.0.1:3306       0.0.0.0:*          users:(("mariadb",pid=30056,fd=23))
tcp    LISTEN 0       128        0.0.0.0:1222        0.0.0.0:*          users:(("sshd",pid=29121,fd=3))
tcp    LISTEN 0       200       127.0.0.1:5432       0.0.0.0:*          users:(("postgres",pid=32479,fd=7))
tcp    LISTEN 0     4096       127.0.0.54:53       0.0.0.0:*          users:(("systemd-resolve",pid=9942,fd=17))
tcp    LISTEN 0     4096     127.0.0.53%lo:53     0.0.0.0:*          users:(("systemd-resolve",pid=9942,fd=15))
tcp    LISTEN 0       128        [::]:1222          [::]:*             users:(("sshd",pid=29121,fd=4))
tcp    LISTEN 0       200        [::1]:5432         [::]:*             users:(("postgres",pid=32479,fd=6))
```

Ver firewall

sudo ufw status verbose

```
seguridad@eqnaranja:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
2222/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
22/tcp DENY IN Anywhere
1222/tcp ALLOW IN Anywhere
2222/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
443/tcp (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) DENY IN Anywhere (v6)
1222/tcp (v6) ALLOW IN Anywhere (v6)
```

Revisar logs SSH

sudo cat /var/log/auth.log

```
seguridad@eqnaranja:~$ sudo cat /var/log/auth.log
2026-05-28T20:23:16.740660+00:00 eqnaranja systemd-logind[838]: New session 149 of user seguridad.
2026-05-28T20:24:04.185841+00:00 eqnaranja sshd[32961]: Received disconnect from 187.194.208.110 port 62929:11: disconnected by user
2026-05-28T20:24:04.186263+00:00 eqnaranja sshd[32961]: Disconnected from user seguridad 187.194.208.110 port 62929
2026-05-28T20:24:04.187196+00:00 eqnaranja sshd[32906]: pam_unix(sshd:session): session closed for user seguridad
2026-05-28T20:24:04.199991+00:00 eqnaranja systemd-logind[838]: Session 141 logged out. Waiting for processes to exit.
2026-05-28T20:24:04.203418+00:00 eqnaranja systemd-logind[838]: Removed session 141.
2026-05-28T20:24:06.236751+00:00 eqnaranja sshd[33642]: Received disconnect from 187.194.208.110 port 65225:11: disconnected by user
2026-05-28T20:24:06.237014+00:00 eqnaranja sshd[33642]: Disconnected from user seguridad 187.194.208.110 port 65225
2026-05-28T20:24:06.239767+00:00 eqnaranja sshd[33544]: pam_unix(sshd:session): session closed for user seguridad
2026-05-28T20:24:06.251159+00:00 eqnaranja systemd-logind[838]: Session 149 logged out. Waiting for processes to exit.
2026-05-28T20:24:06.254793+00:00 eqnaranja systemd-logind[838]: Removed session 149.
2026-05-28T20:24:38.283558+00:00 eqnaranja sshd(pam_google_authenticator)[33656]: Accepted google_authenticator for seguridad
2026-05-28T20:24:38.385978+00:00 eqnaranja sshd[33654]: Accepted keyboard-interactive/pam for seguridad from 187.194.208.110 port 63769 ssh2
2026-05-28T20:24:38.390067+00:00 eqnaranja sshd[33654]: pam_unix(sshd:session): session opened for user seguridad(uid=1000) by seguridad(uid=0)
2026-05-28T20:24:38.400569+00:00 eqnaranja systemd-logind[838]: New session 150 of user seguridad.
2026-05-28T20:25:01.292087+00:00 eqnaranja CRON[33721]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-28T20:25:01.298337+00:00 eqnaranja CRON[33721]: pam_unix(cron:session): session closed for user root
2026-05-28T20:25:21.561529+00:00 eqnaranja CRON[33503]: pam_unix(cron:session): session closed for user root
2026-05-28T20:25:53.387406+00:00 eqnaranja sshd(pam_google_authenticator)[33719]: Accepted google_authenticator for seguridad
2026-05-28T20:25:53.464779+00:00 eqnaranja sshd[33717]: Accepted keyboard-interactive/pam for seguridad from 131.196.245.181 port 49773 ssh2
2026-05-28T20:25:53.468597+00:00 eqnaranja sshd[33717]: pam_unix(sshd:session): session opened for user seguridad(uid=1000) by seguridad(uid=0)
2026-05-28T20:25:53.475682+00:00 eqnaranja systemd-logind[838]: New session 152 of user seguridad.
2026-05-28T20:26:50.631545+00:00 eqnaranja sshd[33838]: Connection reset by authenticating user eqnaranja 187.194.208.110 port 50393 [preauth]
2026-05-28T20:28:10.777877+00:00 eqnaranja sshd[33828]: Received disconnect from 131.196.245.181 port 49773:11: disconnected by user
2026-05-28T20:28:10.783505+00:00 eqnaranja sshd[33828]: Disconnected from user seguridad 131.196.245.181 port 49773
2026-05-28T20:28:10.784597+00:00 eqnaranja sshd[33717]: pam_unix(sshd:session): session closed for user seguridad
2026-05-28T20:28:10.795235+00:00 eqnaranja systemd-logind[838]: Session 152 logged out. Waiting for processes to exit.
2026-05-28T20:28:10.798512+00:00 eqnaranja systemd-logind[838]: Removed session 152.
2026-05-28T20:30:38.798306+00:00 eqnaranja sshd(pam_google_authenticator)[33847]: Accepted google_authenticator for seguridad
2026-05-28T20:30:38.941190+00:00 eqnaranja sshd[33845]: Accepted keyboard-interactive/pam for seguridad from 131.196.245.181 port 49791 ssh2
2026-05-28T20:30:38.944187+00:00 eqnaranja sshd[33845]: pam_unix(sshd:session): session opened for user seguridad(uid=1000) by seguridad(uid=0)
2026-05-28T20:30:38.949924+00:00 eqnaranja systemd-logind[838]: New session 153 of user seguridad.
2026-05-28T20:32:02.297233+00:00 eqnaranja sudo: seguridad : TTY=pts/10 ; PWD=/home/seguridad ; USER=root ; COMMAND=/usr/bin/ss -tulnp
2026-05-28T20:32:02.299938+00:00 eqnaranja sudo: pam_unix(sudo:session): session opened for user root(uid=0) by seguridad(uid=1000)
2026-05-28T20:32:02.317444+00:00 eqnaranja sudo: pam_unix(sudo:session): session closed for user root
2026-05-28T20:32:53.237840+00:00 eqnaranja sudo: seguridad : TTY=pts/10 ; PWD=/home/seguridad ; USER=root ; COMMAND=/usr/sbin/ufw status verbose
2026-05-28T20:32:53.243175+00:00 eqnaranja sudo: pam_unix(sudo:session): session opened for user root(uid=0) by seguridad(uid=1000)
2026-05-28T20:32:53.416561+00:00 eqnaranja sudo: pam_unix(sudo:session): session closed for user root
2026-05-28T20:33:15.488898+00:00 eqnaranja sudo: seguridad : TTY=pts/10 ; PWD=/home/seguridad ; USER=root ; COMMAND=/usr/bin/cat /var/log/auth.log
2026-05-28T20:33:15.492518+00:00 eqnaranja sudo: pam_unix(sudo:session): session opened for user root(uid=0) by seguridad(uid=1000)
seguridad@eqnaranja:~$
```

Ver Fail2Ban

```
sudo fail2ban-client status sshd
```

```
seguridad@eqnaranja:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    12
|   `-- File list:      /var/log/auth.log
`-- Actions
    |- Currently banned: 0
    |- Total banned:    2
    `-- Banned IP list:
```

CONCLUSIÓN

El hardening de GNU/Linux es fundamental para proteger servidores contra ataques y accesos no autorizados. Mediante configuraciones adecuadas de SSH, firewall, autenticación multifactor, usuarios, bases de datos y respaldos automáticos, se incrementa considerablemente la seguridad del sistema y se garantiza la integridad de la información.